Portfolio
cass54.github.io

RAPHAËL CASSARD

MSc Student Engineer @ ENSEEIHT — RF Electronics, Antennas & Microwave Systems

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EDUCATION

- **MSc Engineering Degree 3EA-SysCom** — Sep 2024 - Present
  ENSEEIHT (Toulouse INP), France. Microwave Circuits, Electromagnetic Modeling, Planar & Aperture Antennas, MMIC, MEMS, Radar, EMC.
- **Synopsys Bootcamp — ASIC Design & CPU Architecture (Verilog)** — Jul 2026
  One-week intensive training: Synopsys EDA flow (Design Compiler, VCS, ICC2) & Verilog CPU/SRAM design.
- **Intensive Preparatory Classes for Grandes Écoles (CPGE)** — Sep 2022 - Jul 2024
  Lycée Henri Poincaré, Nancy, France. PCSI - PC* track: Intensive studies in Physics, Chemistry & Engineering.
- **Scientific Baccalaureate (High School Diploma)** — 2019 - 2022
  Lycée Marquette, Pont-à-Mousson, France. Highest Honors (Mention Très Bien) — Math & Physics majors, Advanced Math option, European English Section.

PROFESSIONAL EXPERIENCES & INTERNSHIPS

- **Research Intern — 4.0 GHz Antenna Design, CNC Prototyping & RF Testing** — Jun - Aug 2026
  Universidad San Francisco de Quito (USFQ) — Cumbayá / Quito, Ecuador.
  - Analytical modeling and 3D simulation in Ansys HFSS of a wideband 4.0 GHz microstrip patch antenna (capacitive gap-coupling with parasitic resonator) on FR-4 for SDR localization (ToA/AoA).
  - Laboratory fabrication via CNC (KiCad, FlatCAM), $FeCl_3$ chemical etching, SMA soldering.
  - RF network analyzer measurements: measured 370 MHz (9.25%) -10 dB bandwidth (target ≥ 200 MHz); designed and assembled a custom 2D testbench for far-field radiation patterns.
- **Industrial Maintenance Mechanic** — Jun - Aug 2025
  Rénovation Machines Industrielles (RMI), Nomeny and various industrial sites, France.
  - Preventive maintenance and mechanical intervention on diverse industrial machinery fleets (in workshop and on-site).
  - Developed strong technical autonomy, rigor, and adaptability in demanding industrial environments.
- **Biomedical Maintenance Operator** — Aug - Sep 2023
  Quipment, Nancy, France.
  - Preventive maintenance, quality control, and configuration of medical devices according to clinical standards.

PROJECTS

- **5.2 GHz Microwave Amplifier & Radio Transmitter (ADS / Momentum)** — Year 2
  Complete design of a 5.2 GHz high-frequency amplifier for a radio transmitter. Component selection, impedance matching, RF PCB layout design, and electromagnetic simulation using Keysight ADS Momentum.
- **Horn & Patch Antennas for 10 GHz Microwave Link (HFSS / MATLAB)** — Year 2
  Design and simulation in Ansys HFSS of a horn antenna and a microstrip patch on Rogers substrate at 10 GHz. Radiation pattern extraction and link budget evaluation in MATLAB.
- **Electromagnetic Waveguide Structure Modeling (HFSS / MATLAB)** — Year 2
  Analytical derivation of the dispersion equation via equivalent circuit method. Numerical solving in MATLAB and dispersion curve validation against Ansys HFSS 3D simulations.
- **Optoelectronic FM Audio Link — Transmitter & Receiver (PSpice)** — Year 2
  Design and assembly of a complete optical FM audio link: VCO modulation, PLL demodulation, selective Cascode amplifier stage, and active Sallen-Key filtering.

SKILLS & TOOLS

- **Technical Domains:** RF Electronics, Antenna Design, Electromagnetic Modeling, Microwave Circuits (MMIC, MEMS), Analog & Digital Electronics, Telecoms & Signal Processing.
- **Software & EDA Tools:** Synopsys (1-week training), ADS (RF & Momentum), ANSYS HFSS, Orcad PSpice, Vivado, Quartus, MPLAB X • **Languages:** VHDL, Verilog, C/C++, Python, MATLAB & Simulink.
- **Soft Skills:** Leadership (70-member team coordination) • Social Skills & Team Player • Event Logistics • Technical Autonomy & Adaptability • Ethics & CSR.
- **Languages:** French (Native) • English (B2+ Certified Cambridge) • Italian (A2 - Intermediate).

COMMUNITY INVOLVEMENT & INTERESTS

- **Vice-President of Student List (2024 – 2025):** Coordination of a 70-member team, budget management (> €15k), and organization of major school events.
- **Afterwork Manager — ENSEEIHT Student Association (2025 – 2026):** Weekly event management and logistics for the student union bar ("Foyer").
- **Interests:** Competitive Swimming (Regional Level) • Electric Guitar (8 years of practice).